ARYAN TANEJA

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Education

University of Illinois, Urbana-Champaign (UIUC) **Aug 2024 - May 2026**
M.S in Computer Science (Advisor - Prof. Deepak Vasisht) *GPA : 4.0/4.0*
- *Focus: AI Infrastructure, Systems, and Networking*

IIIT Delhi **July 2019 – Jun 2023**
Bachelor of Technology in Computer Science and Engineering (Rank 3) *GPA: 9.32/10.0*

Work Experiences

Netflix: Software Engineer (Distributed Systems) **June'26 – present**
Open Connect Foundations and Infrastructure *Los Gatos, CA*
- Developed server-side efficiency optimizations to improve high-throughput infrastructure (OCAs) handling **O(Million) RPMs by 13% mean (upto 42% peak)**
- Developed headless AI agents ingesting automated telemetry pipelines for A/B testing, enabling autonomous bottleneck diagnosis and minimizing system latency at scale.
- Tuned packet pacing and capacity estimation to improve throughput efficiency by **23% (upto 48%)** on dynamic paths

Netflix: Summer Intern **May'25 – Aug'25**
Mentors: Dr.Renata Teixeira, Dr. Reese Enghardt | Serving Systems, Transport Protocols *Los Gatos, CA*
- Applied data modeling and AI tooling to investigate path properties and optimize the OCA TCP stack, developing a cross-platform (server/client) firmware solution that enhanced reliability and QoE for **O(Million) users**.

Microsoft: Research Fellow **Feb'23 – Aug'24**
Mentor: Dr.Venkat Padmanabhan | Systems for LLMs *Bangalore, India*
- Optimized vllm-based systems to serve, schedule, migrate and load balance **LLM inference requests** for cost and GPU throughput efficiency (**upto 34%**). Implemented KV cache management & migration module for efficient global routing
- Implemented a **kernel level service** which used real-time network to improve throughput by **120%** by optimising congestion control for dynamic paths.
- Prototyped Agentic AI pipelines using frontier models to recursively implement research code, successfully extracting and deploying logic from over 50 technical RFCs and research papers

Microsoft: Research Intern **Feb'22 – Jun'22**
Mentor: Dr. Debopam Bhattacherjee | Network Performance Optimisation & Analysis *Bangalore, India*
- Used **Sentiment Analysis, Computer Vision, implicit & explicit user signals** like presence, activity, engagement for User and **Network Performance Modelling** for Starlink, XBox-forza (Azure Cloud) and Microsoft Teams

MathWorks: SDE Intern **Jun'22 – Aug'22**
Full-stack Development *Bangalore, India*
- Implemented a code violation tracking system into production which improved the compilation efficiency by **196%**

Technical Skills

Languages: Java, C/C++, Python, Bash
AI Frameworks & Tooling: vLLM, SGLang, Ollama, MCPs, LangGraph, LlamaIndex, Ray, Wireshark, perf, Docker, Git, AWS, MySQL, gRPC

Selected Projects

VoltanaLLM: Feedback-Driven Frequency and Routing for Efficient LLM Serving **Jan'25 – May'26**
vLLM, SGLang | Efficient AI Infra blog, code, paper accepted at ISC High Performance'26 *Prof. Minjia Zhang, UIUC*
- Designed a serving system that leverages feedback-driven GPU frequency control and state-space routing on disaggregated architectures. By dynamically optimizing hardware states during LLM inference, the system reduced A100 GPU energy consumption by **up to 36.3%** while strictly maintaining latency-based Service Level Objectives (SLOs).

LEOScope : an open source global-testbed for large-scale telemetry and experimentation **Jan'23 – Aug'24**
Docker | grpc | FastAPI | ReactJS | Node.JS | website, code, paper *Microsoft Research*
- Built (full-stack), deployed and maintaining LEOScope, (public release) with a cloud orchestrator and geodistributed node machines to run **containerised dockerised** systems+networking jobs for Starlink at scale.

Optimising Video Streaming with Edge Computing **May'21 – Jul'23**
Ffmpeg | Yolo/Open-CV | Jetson Nano | Patent Accepted, USENIX ATC'24 (CA, USA), COMSNETS *IIIT-Delhi*
- Engineered a real-time edge video surveillance system using tile sampling and HEVC/H.265 bitrate fluctuations, achieving **92%** object detection accuracy and reducing data transmission by **40%**.